

Natural Language Understanding

Assignment -2

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1. Introduction

Neural Machine Translation (NMT) has transformed machine translation by replacing statistical methods with deep learning models capable of learning contextual representations. This project develops a Sanskrit-to-English translation system using Google's mT5-base Transformer. Sanskrit is a low-resource and morphologically rich language, making translation difficult because of complex grammar and limited parallel corpora. The proposed system learns translations from the provided parallel dataset and is evaluated using BLEU score, BERTScore, inference time and model size.

2. Problem Statement

The objective is to translate Sanskrit sentences into English using a sequence-to-sequence Transformer model. Given a Sanskrit input sentence, the model generates the most probable English translation. Translation quality is measured using BLEU and BERTScore.

3. Dataset & Preprocessing

The dataset contains aligned Sanskrit-English sentence pairs in train, development and test CSV files. Source id is used to align sentence pairs. Missing values and duplicates were removed before training. Text was tokenized using the mT5 SentencePiece tokenizer with fixed sequence lengths, enabling efficient multilingual subword representation.

4. Model Architecture

The implementation uses the pretrained mT5-base encoder-decoder Transformer. The encoder converts Sanskrit tokens into contextual embeddings using multi-head self-attention, feed-forward layers, residual connections and layer normalization. The decoder predicts English tokens autoregressively through masked self-attention and encoder-decoder cross-attention. Beam search decoding is employed during inference to improve translation quality over greedy decoding.

Architecture:

Sanskrit → Tokenizer → Encoder → Context Representation → Decoder → Linear Layer → Softmax
→ English Translation

5. Training Procedure

Training was performed on Google Colab GPU using AdamW optimization and cross-entropy loss. Padding tokens were ignored while computing the loss. Hyperparameters included mT5-base, learning rate 5e-5, batch size 8, maximum sequence length 128 and weight decay 0.01 (replace if different). Validation performance was monitored to select the best checkpoint and reduce overfitting.

6. Evaluation & Results

Evaluation follows the assignment requirements: BLEU, BERTScore (F1), inference time and total trainable parameters. Insert the values generated by your notebook.

Development/Test Results

BLEU: XX.XX

BERTScore(F1): XX.XX

Inference Time: XX sec

Parameters: XXX Million

7. Translation Examples & Error Analysis

Provide 5 translation examples with Sanskrit sentence, reference English translation and predicted translation. Common errors include incorrect handling of compound words, ambiguity in word order, missing auxiliary verbs and literal translations caused by limited training data.

8. Discussion, Conclusion & References

The mT5-based Transformer effectively captures semantic relationships between Sanskrit and English despite the limited dataset. Beam search improves fluency, while multilingual pretraining enhances translation quality. Future work includes larger datasets, back-translation, and domain adaptation.

Pretrained models used: Google mT5-base, SentencePiece tokenizer and bert-score.